

Maths-1 - Combinatorics (Part 2) Work-along Sheet

Name: _____; Roll Number: _____

Lecture Batch: _____; Lab Batch: _____

1. Find the number of ways 10 people can be arranged in a row such that

i) a specific pair is always together

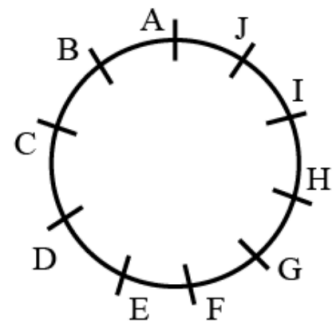
ii) a specific pair is never together

2. Find the number of ways n people can be arranged linearly such that

i) a specific pair is always together

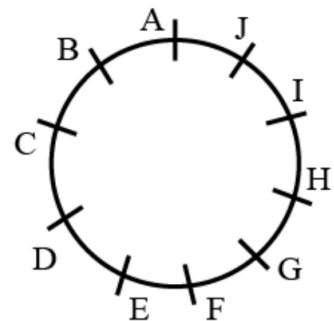
ii) a specific pair is never together

3. Find the number of ways 10 people can be arranged in a circle such that
- i) a specific pair is always together



- ii) a specific pair is never together

4. Find the number of ways n people can be arranged in a circle such that
- i) a specific pair is always together



- ii) a specific pair is never together

5. Find the number of ways 4 dogs and 4 cats can be arranged linearly such that
i) 4 cats are together

ii) no 2 cats are together

iii) dogs and cats are arranged simultaneously (dogs and cats must alternate, i.e., no two dogs are adjacent and no two cats are adjacent)

6. Find the number of ways 4 dogs and 4 cats can be arranged in a circle such that
i) 4 cats are together

ii) no 2 cats are together

iii) dogs and cats are arranged simultaneously

Note: Combination means selections of objects where order doesn't matter.

Theorem:

- The number of r -combinations of a set with n elements, where n is a nonnegative integer and r is an integer with $0 \leq r \leq n$, is given by:

$$C(n, r) = \frac{n!}{r!(n - r)!}$$

Proof using Product Rule:

- Consider $P(n, r)$: the number of r -permutations of n elements.
- First choose an r -combination, then arrange it:

$$P(n, r) =$$

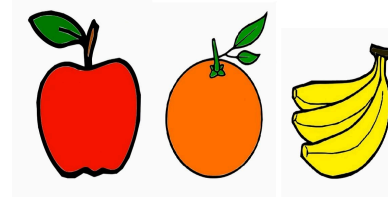
$$\Rightarrow C(n, r) = \text{————} = \text{————} = \text{————}$$

- To choose r objects from n , think of partitioning the set of n objects into 2 groups:
 - One group of r selected objects
 - One group of $n - r$ not selected objects
- Now, we can think of arrangements of n objects in which a set of r objects is identical and another set of $(n-r)$ objects is identical.

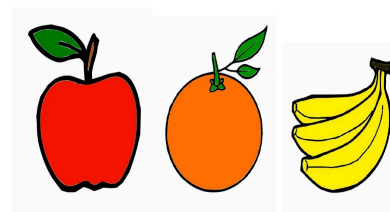
- This is the formula for combinations, written as:

$$\binom{n}{r} = \frac{n!}{r!(n - r)!}$$

7. In how many ways can you select 1 item from a set of 3 distinct items?



8. In how many ways can you select 2 items from a set of 3 distinct items?



9. In how many ways can you select 2 books from a set of 4 books?



10. How many ways can you choose 3 fruits from a basket containing apples, bananas, and oranges (assuming an unlimited supply of each)?

Number of Apples	Number of Bananas	Number of Oranges



The **stars and bars** method (also known as "balls and urns" or "sticks and stones") is a powerful counting technique in mathematics used to determine the number of ways to distribute indistinguishable objects into distinguishable bins or to find the number of non-negative integer solutions to an equation of the form $x_1 + x_2 + \dots + x_k = n$.

Representation: Indistinguishable objects are represented by "stars" (*), and "bars" (|) are used as dividers to separate the objects into different groups or "bins".

Now, let's represent the given problem using the Stars and Bars pattern.

1 apple, 1 banana, 1 orange can be represented as * | * | *

2 apples, 0 bananas, 1 orange can be represented as ** | | *

0 apples, 0 bananas, 3 oranges can be represented as _____

_____ **apples**, _____ **bananas**, _____ **oranges** can be represented as | * * * |

_____ *'s and _____ |'s are needed to represent the scenario mentioned in the above problem. It means that we need _____ slots to represent these *'s and |'s.

Case 1: Select *'s

Select _____ slots for _____ *'s out of total _____ slots.

So, _____ |'s are automatically placed in the remaining slots.

This can be done in _____ ways.

Case 2: Select |'s

Select _____ slots for _____ |'s out of total _____ slots.

So, _____ *'s are automatically placed in the remaining slots.

This can be done in _____ ways.

So, the logic is to select either stars or bars.

Q. What is your observation about the Case 1 and Case 2 answers?

$${}^nC_r =$$

$${}^nC_{n-r} =$$

We observe that _____ = _____.

Intuitive Example:

Imagine you have 5 fruits (say, apples), and you want to select 2 of them. The number of ways to choose 2 apples from 5 is the same as the number of ways to leave out 3 apples (since choosing 2 apples means you're automatically leaving out 3). Both are equivalent problems. So, _____ = _____.

11. How many ways can you choose 5 fruits from a basket containing apples, bananas, and oranges (assuming an unlimited supply of each)?

12. How many ways can you select 10 chocolates from 4 different varieties, where the selection of each variety is unlimited?



13. In how many ways can you distribute 5 identical candies among 4 children?



14. How many lines can be made using 'n' distinct points?

- **Case 1:** No 3 points are collinear

We need to select _____ points out of _____ points in order to make a line. We can select _____ points out of _____ points in _____ ways.

- **Case 2:** Some points are collinear, i.e., 'k' points out of 'n' points are collinear

Using k points, we can make _____ lines but we are able to make only _____ line because _____ lines are collinear.

- **Case 3:** Special case (All points are collinear)

- **Case 4:** All points are the same point

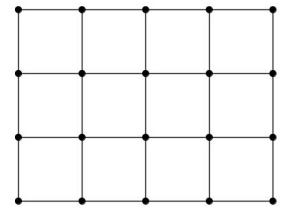
15. How many lines can be made using 15 distinct points such that a specific set of 3 distinct points is collinear and another set of 5 distinct points is also collinear?

16. How many triangles can be made using 'n' distinct points such that no three points are collinear?

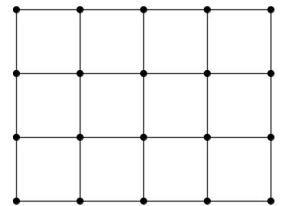
17. How many triangles can be made using 'n' distinct points such that 'k' points are collinear?

18. How many triangles can be made using 15 distinct points such that a specific set of 3 distinct points is collinear and another set of 5 distinct points is also collinear?

19. How many rectangles are present in 3 x 4 grid?



20. How many squares are present in 3 x 4 grid?

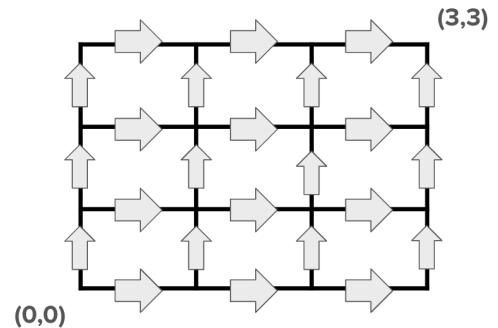


21. How many rectangles are present in 8 x 8 grid?

22. How many squares are present in 8 x 8 grid?

Increasing Path Problem

23. How many paths are possible from (0,0) to (3,3) such that at every step we can move 1 unit right or 1 unit up only?



24. How many paths are possible from (0,0) to (4,5) such that at every step we can move 1 unit right or 1 unit up only?

Number of paths from (x_1, y_1) to (x_2, y_2) on the Cartesian plane such that at each step we can move 1 unit right or 1 unit up only

$$= {}^{(x_2-x_1)+(y_2-y_1)}C_{(x_2-x_1)}$$

$$= {}^{(x_2-x_1)+(y_2-y_1)}C_{(y_2-y_1)}$$

25. At every step, we can move 1 unit right or 1 unit up only.

Find the number of increasing paths from

i) (0,0) to (8,8)

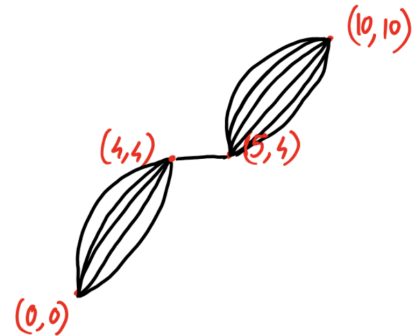
ii) (0,0) to (10,10)

iii) (4,4) to (5,4)

iv) (5,4) to (10,10)

26. How many paths are possible from $(0,0)$ to $(10,10)$, not using $(4,4)$ to $(5,4)$, such that at every step we can move 1 unit right or 1 unit up only?

Increasing paths from
 $(0,0)$ to $(10,10)$, not using $(4,4)$ to $(5,4)$ = -



27. How many paths are possible from $(0,0)$ to $(8,8)$, not using $(2,3)$ to $(5,6)$, such that at every step we can move 1 unit right or 1 unit up only?

Increasing paths from
 $(0,0)$ to $(8,8)$, not using $(2,3)$ to $(5,6)$ = -